

PAPER_665: UQFF Hawking Suppression Equations

Subtitle: Derives the chain of multiplicative suppression factors S_1 , S_2 , S_3 that reduce Hawking luminosity in UQFF, with sensitivity analysis. **Module:** UQFFSuppressionEquationsHawking

Session: Session 172

Date: April 2, 2026

Version: v5.29

Status: Complete — CP4 #249 | UQFF Session 172

Abstract

Three UQFF suppression factors reduce Hawking radiation luminosity multiplicatively. We derive each factor analytically, combine them into a total suppression S_{total} , and perform sensitivity sweeps over the aether density ratio.

1. Suppression Factors

S1 — Negentropic Modulation (affects T_H only)

$$S_1 = 1 + f_{TRZ} = 1.1$$

S2 — [UA]/[SCm] Density Ratio

$$S_2 = 1 - \frac{\rho_{SCm}}{\rho_{UA}} = 0.9$$

S3 — Magnetic String Exponential

$$S_3 = \exp\left(-\frac{U_m}{k_B T_H}\right)$$

2. Modified Quantities

$$T_{UQFF} = T_H \cdot S_1 \cdot S_2$$

$$L_{UQFF} = L_H \cdot S_2 \cdot S_3$$

$$\frac{dT_{UQFF}}{dM} = \frac{dT_H}{dM} \cdot S_1 \cdot S_2$$

3. Total Suppression

$$S_{total} = S_1 \cdot S_2 \cdot S_3$$

For $U_m/k_{BT_H} = 1$: $S_{total} \approx 1.1 \times 0.9 \times 0.368 \approx \mathbf{0.364}$

Hawking luminosity reduced to $\sim 36\%$ of GR prediction.

4. Sensitivity Sweep

As ρ_{UA}/ρ_{SCm} varies from 2 to 20: S_2 ranges 0.5→0.95, strongly affecting L_{UQFF} .

5. C++ Module

UQFFSuppressionEquationsHawking.h / .cpp — Session 172 CP4 #249 — UQFFSuppressionEquationsHawkingCalculator

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